

Custodial paper map — where each formula comes from, and what each paper compares

Which papers gave us *formulas* (top group) vs. *comparison/validation numbers* (bottom group). Detail:
docs/CUSTODIAL_PROVENANCE.md.

Papers we took custodial formulas FROM

CGHNP — Flavor in minimal RS, I (Casagrande, Goertz, Haisch, Neubert, Pfoh; arXiv:0807.4937)

Formulas we took: minimal ΔT (Eq. 147); **custodial ΔT (Eq. 153)** — the exact closed form our code uses; $Z \rightarrow b\bar{b}$ tree g_L^b, g_R^b (Eq. 170).

Computes (custodial vs. non): minimal-RS S, T, U and $Z \rightarrow b\bar{b}$; presents custodial $SU(2)_R$ as “cure #4” for T — minimal $T \Rightarrow M_{KK} > 4 \text{ TeV}$ vs. custodial (S-limited) $\Rightarrow 2.4 \text{ TeV}$.

ADMS — RS1, Custodial Isospin and Precision Tests (Agashe, Delgado, May, Sundrum; hep-ph/0308036)

Formulas we took: the custodial- T mechanism ($SU(2)_R$ kills the $\log(k\pi r_c)$ enhancement); the origin distilled into CGHNP Eq. 153 (analytic; no closed form copied directly).

Computes (custodial vs. non): T (protected $\rightarrow \sim 3 \text{ TeV}$ floor from S), S (unprotected), Zb_L ; the custodial vs. log-enhanced-minimal T contrast.

ACDP — A custodial symmetry for $Zb\bar{b}$ (Agashe, Contino, Da Rold, Pomarol; hep-ph/0605341)

Formulas we took: the P_{LR} protection $\delta g_L^b \rightarrow 0$ (Eqs. 3–12); the b_R representation map (Table 1) $\Rightarrow$ our $g_R^b=0$ choice ($b_R \in (1, 1)_{-1/3}$).

Computes (custodial vs. non): δg_L^b (protected) and δg_R^b (rep-dependent sign) custodial vs. minimal; the A_{FB}^b anomaly.

CPSW — Light KK / EW constraints with custodial $SU(2)$ (Carena, Pontón, Santiago, Wagner; hep-ph/0607106 + hep-ph/0701055)

Formulas we took: the top-partner one-loop δg_L^b (Eqs. 14–15) and ΔT (Eq. 13, with T_{top} Eq. 35).

Computes (custodial vs. non): custodial T (bidoublet $-T$, singlet $+T$), the loop δg_L^b , and S ; custodial gauge-KK floor $\sim 2.5\text{--}3 \text{ TeV}$.

Papers used for COMPARISON / VALIDATION (no formula taken)

CFW — The Flavor of the Composite pGB Higgs (Csáki, Falkowski, Weiler; arXiv:0804.1954)

Formulas we took: none.

Computes (custodial vs. non): anarchic ε_K (KK-gluon $\Rightarrow M_{KK} \gtrsim 21 \text{ TeV}$ RS / 33 TeV GHU) *assuming* custodial EW ($\sim 3 \text{ TeV}$) — the explicit EW-vs-flavor split we reproduce.

Blanke et al. — $\Delta F = 2$ with custodial protection (Blanke, Buras, Duling, Gori, Weiler; arXiv:0809.1073)

Formulas we took: none.

Computes (custodial vs. non): the full $\Delta F = 2$ set *inside the custodial model* $\Rightarrow \varepsilon_K$ wall *stays* $M_{KK} \gtrsim 18\text{--}30 \text{ TeV}$ — the proof that custodial does not touch flavor.

Bauer II — Flavor in RS, II (Bauer, Casagrande, Haisch, Neubert; arXiv:0912.1625)

Formulas we took: none (for custodial).

Computes (custodial vs. non): anarchic minimal ε_K (our Lane-A baseline, $M_g^{(1)} \gtrsim 10 \text{ TeV}$); notes Zb_L is removable by custodial.

One-line summary. We take the *leading* custodial formulas from CGHNP (the T closed forms), ADMS (the T mechanism), ACDP (P_{LR} $Zb\bar{b}$), and CPSW (the loops); CFW, Blanke, and Bauer supply the comparison numbers that show custodial is an *electroweak* cure (S, T, U and $Z \rightarrow b\bar{b}$ to a few TeV), not a flavor cure (the ε_K wall stays at tens of TeV).